Data preprocessing -

ML → Model training

→ pandas ingest dataFrame → df

features target

→ dtype match according to the data in the columns

→ no null values (Remove) → mean, median, std

→ Scaling (4 columns in same scaling)

→ train, test

profiling of data.

Select * from ~~horse power~~ mpg where horsepower = '?'

→ df [df['horsepower'] == '?']

inject into df properly → read_csv

dtypes → mismatch - Solve

isna → ✓ (SI) mean, median, std

isna → ✓ (ST) mean, median, std

1) standard scalar
2) minmax scalar
3) Robust scalar
4) Normaliser

Age	salary (K)
20	30
25	50
30	60
35	70
40	90
(X)	(x)

Age: 20, 25, 30, 35, 40

$x' = (x - \mu)/\sigma$

$\mu = (20 + 25 + 30 + 35 + 40)/5 = 30$ ✓

$\sigma_{age} = \sqrt{(-10)^2 + (-5)^2 + 0^2 + (5)^2 + 10^2}$

$= \sqrt{100 + 25 + 0 + 25 + 100}$

$= \sqrt{250/5} = \sqrt{50} = 7.0711$

Age	$x-\mu$	$\div \sigma$	$x' = (x-\mu)/\sigma$
20	-10	7.0711	$= -10/7.0711 = -1.42$
25	-5	7.0711	$= -0.7131$
30	0	7.0711	0
35	5	7.0711	0.7131
40	10	7.0711	1.42

2) <u>MinMaxScalar</u>: "squeeze everything into [0,1]"

Formula: $x' = (x-\min)/(\max-\min)$

in bounded-range [-1, 1]

image pixel (R, G, B) → [0, 1]

0 0 0

255 45 45

Age	Salary
20	30
25	50
30	60
35	70
40	90

Age – column → x = 20, 25, 30, 35, 40

min = 20 max = 40

Age (x)	x − min	max − min	x − min / max − min	x'
20	0	20	0/20	0
25	5	20	5/20	0.25
30	10	20	10/20	0.5
35	15	20	15/20	0.75
40	20	20	20/20	1

(8) Robust Scaler: "center on median spread IQR"

Formula = $x' = (x - \text{median}) / IQR$

where $IQR = Q_3 - Q_1$

Age sorted = 20, 25, 30, 35, 40

(Q_1 = 25, median = 30, Q_3 = 35)

median = 30, $Q_1 = 25$, $Q_3 = 35$, $IQR = 10$

Salary sorted = 30, 50, 60, 70, 90

(Q_1 = 50, median = 60, Q_3 = 70)

median = 60, $Q_1 = 50$, $Q_3 = 70$ $IQR = 20$

Age (x)	x − 30/10	x'	salary (x)	x − 60/20	x'
20	−10/10	−1	30	−30/20	−1.5
25	−5/10	−0.5	50	−10/20	−0.5
30	0/10	0	60	0/20	0
35	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

30	0/10	0	60	0/20	0
35	5/10	0.5	70	10/20	0.5
40	10/10	1	90	30/20	1.5

$2^0 = 1 + 2^{1-1} = 2^1 \times 2^0$

$= 2/2 + 1$ TOR

(4) Normalizer: "make each row a unit vector"

per-row, per-sample

$L_2 = x' = x / \|x\|_2$

$x_i = x_i / \|x\|_2$ where $\|x\|_2 = \sqrt{(\Sigma x_i^2)}$

36.055

$\sqrt{20^2 + 30^2} = \sqrt{1300}$

$= 36.055$

Age	Salary
20/36.055	30/36.055
25/55.90	50/55.90
30/67.05	60/67.05
35/78.26	70/78.26
40/89.44	90/89.44

$\sqrt{25^2 + 50^2} = \sqrt{625 + 2500}$

$= \sqrt{3125}$

$\sqrt{30^2 + 60^2} = \sqrt{900 + 3600}$

$= \sqrt{4500}$

$\sqrt{35^2 + 70^2}$

$1225 + 4900$

6100

1600

raw data → preprocessing → Scaling → ...

raw data → preprocessing → Scaling → feature → model

minmax → cheapest

normalizer → —

standardscaler → modelete

Robust scaler → explain

```
scaler = standscaler()
scaler.fit(X)
X1new = scaler.fit(X)
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
```

X

| x_train | 80%

x_train 80-1

x_test 20

fit_transform(x_test)

(random)